Assignment 4: Customer Churn Prediction with XGBoost

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Problem & Data

The IBM Telco Customer Churn dataset (7,043 customers, 19 features) is used to predict whether a customer will churn. This is a **binary classification** problem with class imbalance: 73% No Churn vs 27% Churn. Features include contract type, tenure, internet service, monthly charges, and various add-on services. `TotalCharges` was converted from string to numeric, and `customerID` was dropped.

Evaluation Metric: Recall

Recall was chosen as the primary metric because the cost of missing a churner (lost revenue, no chance to intervene) far exceeds the cost of a false alarm (offering a retention deal to a loyal customer). With 73/27 class imbalance, accuracy is misleading — a naive "predict No Churn" model achieves 73% accuracy while catching zero churners.

Preprocessing & Models

Numeric features (tenure, MonthlyCharges, TotalCharges, SeniorCitizen) were median-imputed and standardized. Categorical features (15 columns) were one-hot encoded. Data was split 80/20 with stratification.

- **Baseline**: Logistic Regression (`max_iter=1000`)
- **XGBoost**: Tuned via `GridSearchCV` (scoring=recall, 3-fold CV) over `n_estimators`, `max_depth`, `learning_rate`, and `scale_pos_weight`

Best hyperparameters: `n_estimators=100`, `max_depth=3`, `learning_rate=0.01`, `scale_pos_weight=3`

Results

Metric	Logistic Regression	XGBoost
Accuracy	0.8055	0.7111
Precision	0.6572	0.4744
Recall	**0.5588**	**0.8182**
F1-Score	0.6040	0.6006
ROC-AUC	0.8419	0.8368

XGBoost catches **306 of 374 churners** (82%) vs logistic regression's 209 (56%). The tradeoff is lower precision (more false alarms), but for churn prevention this is the right direction. F1 and ROC-AUC are nearly identical, confirming XGBoost isn't worse overall — just tuned to prioritize the minority class via `scale_pos_weight`.

Top Features

Rank	Feature	Importance
1	Contract: Month-to-month	0.562
2	InternetService: Fiber optic	0.092
3	TechSupport: No	0.087
4	OnlineSecurity: No	0.064
5	tenure	0.044

Month-to-month contracts dominate — customers with no commitment churn at far higher rates. **Fiber optic** customers and those **without add-on services** (tech support, online security, online backup) are also high-risk. Short **tenure** and **electronic check** payment further signal churn risk.

Conclusion

XGBoost with `scale_pos_weight=3` significantly outperforms logistic regression on recall while maintaining comparable ROC-AUC. The model identifies actionable retention levers: incentivize longer contracts, bundle add-on services, and target newer customers with retention offers. A limitation of XGBoost is reduced interpretability — feature importances show relative contribution but not directionality, unlike logistic regression's signed coefficients.